

Phylogenetic Distance Estimation for the 6502life Noisy Copy Model

6502life project

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Abstract

We derive closed-form distance formulas for estimating evolutionary divergence between cells in the 6502life cellular automata system, where replication proceeds through a hierarchical noisy copy channel. The noise model operates at three levels—cell, byte, and bit—producing correlated errors within bytes. We show that the per-byte Hamming distance follows a mixture of two binomials, with eigenvalues given by the Krawtchouk polynomials of the underlying binary symmetric channel. We derive maximum likelihood estimators at three levels of resolution and validate them against Monte Carlo simulations.

1 The Noisy Copy Channel

In 6502life, cell replication proceeds via the `copyCellWithNoise` mechanism, which copies $M = 1024$ bytes from a source cell to a destination cell with three levels of noise:

1. **Cell level:** With probability p_{CN} , the entire destination cell is filled with random bytes (all $8M$ bits i.i.d. Bernoulli($1/2$)).
2. **Byte level:** For each byte, with probability p_{BN} (given the cell was not randomized), the byte is replaced by a random byte.
3. **Bit level:** For each bit, with probability $\varepsilon = p_{\text{bit}}$ (given neither cell nor byte noise fired), the bit is replaced by a random bit.

The default parameters are $p_{\text{CN}} = 0$, $p_{\text{BN}} = 0$, $\varepsilon = 0.001$.

1.1 Effective rates

The *per-byte randomization rate* (probability that all 8 bits in a byte are simultaneously randomized):

$$r = p_{\text{CN}} + p_{\text{BN}}(1 - p_{\text{CN}}) \tag{1}$$

The *effective per-bit randomization rate*:

$$\varepsilon_{\text{eff}} = 1 - (1 - p_{\text{CN}})(1 - p_{\text{BN}})(1 - \varepsilon) \tag{2}$$

2 Bit-Level Distance (Binary Jukes–Cantor)

Each bit evolves under a 2-state symmetric Markov chain. Per copy event, a bit is randomized (replaced by a fair coin flip) with probability ε_{eff} , and faithfully copied otherwise.

Proposition 1. *After T copy events on the tree path between two leaves (total evolutionary distance), the probability that a single bit differs is*

$$p_{\text{bit}}(T) = \frac{1}{2}(1 - (1 - \varepsilon_{\text{eff}})^T) \quad (3)$$

Proof. The per-bit pairwise comparison follows a 2-state Markov chain with transition rate $\varepsilon_{\text{eff}}/2$ in each direction (same $\rightarrow$ different and different $\rightarrow$ same), since randomization produces a match with probability $1/2$ regardless of the current state. Starting from “same” at $T = 0$:

$$P(\text{diff after } T) = \frac{1}{2}(1 - (1 - \varepsilon_{\text{eff}})^T)$$

This is the standard binary Jukes–Cantor model. □

Inverting:

$$T = \frac{\log(1 - 2p_{\text{bit}})}{\log(1 - \varepsilon_{\text{eff}})} \quad (4)$$

Saturation: $p_{\text{bit}} \rightarrow 1/2$ as $T \rightarrow \infty$.

3 The 9-State Byte Hamming Model

3.1 Markov chain on Hamming distance

Consider a single byte position. The *pairwise Hamming distance* $h \in \{0, 1, \dots, 8\}$ (number of bit positions where two corresponding bytes differ) evolves as a 9-state Markov chain.

Per copy event on one lineage:

- With probability r : all 8 bits are randomized. Regardless of the current state, each bit independently becomes “same” or “different” with equal probability. The new $h \sim \text{Bin}(8, 1/2)$.
- With probability $1-r$: each of the h “different” bits repairs (becomes “same”) with probability $\varepsilon/2$, and each of the $8-h$ “same” bits breaks (becomes “different”) with probability $\varepsilon/2$.

3.2 Eigenvalues via Krawtchouk polynomials

The transition matrix P for this chain has the structure

$$P = (1 - r)Q + r \mathbf{1} \boldsymbol{\pi}^\top$$

where Q is the bit-level-only transition matrix and $\boldsymbol{\pi} = (\text{Bin}(h; 8, 1/2))_{h=0}^8$ is the stationary distribution.

The matrix Q describes 8 independent binary symmetric channels projected onto their Hamming weight. Its eigenvectors are the Krawtchouk polynomials $K_k(h; 8, 1/2)$ with eigenvalues $\mu_k = (1 - \varepsilon)^k$ for $k = 0, \dots, 8$.

Since $\boldsymbol{\pi}^\top K_k = 0$ for $k \geq 1$ (orthogonality), the eigenvalues of P are:

$$\lambda_0 = 1, \quad \lambda_k = (1 - r)(1 - \varepsilon)^k \quad \text{for } k = 1, \dots, 8 \quad (5)$$

3.3 Closed-form distribution: mixture of two binomials

Theorem 1. *The distribution of per-byte Hamming distance h between two cells separated by T copy events on the tree, starting from $h = 0$, is*

$$\boxed{\pi_T(h) = (1 - s) \text{Bin}(h; 8, \frac{1}{2}) + s \text{Bin}(h; 8, p)} \quad (6)$$

where

$$s = (1 - r)^T \quad (\text{probability of no byte-randomization on the path}) \quad (7)$$

$$p = \frac{1}{2}(1 - (1 - \varepsilon)^T) \quad (\text{per-bit mismatch from bit-noise alone}) \quad (8)$$

Proof. Using the Krawtchouk generating function $\sum_{k=0}^n z^k K_k(h; n) = (1 - z)^h (1 + z)^{n-h}$ and the spectral decomposition starting from $h_0 = 0$ (where $K_k(0; 8) = \binom{8}{k}$):

$$\pi_T(h) = \frac{\binom{8}{h}}{256} \sum_{k=0}^8 \lambda_k^T K_k(h) \quad (9)$$

$$= \frac{\binom{8}{h}}{256} \left[1 + s \left(\sum_{k=0}^8 z^k K_k(h) - 1 \right) \right] \quad \text{where } z = (1 - \varepsilon)^T \quad (10)$$

$$= \frac{\binom{8}{h}}{256} [1 - s + s(1 - z)^h (1 + z)^{8-h}] \quad (11)$$

Setting $p = (1 - z)/2$:

$$\pi_T(h) = (1 - s) \frac{\binom{8}{h}}{256} + s \binom{8}{h} p^h (1 - p)^{8-h}$$

which is the mixture (6). $\square$

Interpretation. With probability s , the byte was never hit by byte-level or cell-level noise, so h arises purely from bit-noise accumulation: $h \sim \text{Bin}(8, p)$. With probability $1 - s$, the byte was randomized at some point, erasing all signal: $h \sim \text{Bin}(8, 1/2)$.

3.4 Moments

The mean of π_T is

$$\mathbb{E}[h] = (1 - s) \cdot 4 + s \cdot 8p = 4(1 - (1 - \varepsilon_{\text{eff}})^T) = 8p_{\text{bit}}(T)$$

which is identical to the bit-level formula. The variance, however, is inflated relative to a single binomial when $r > 0$, due to the mixture.

3.5 Special cases

Default parameters ($r = 0, s = 1$): The mixture collapses to a single $\text{Bin}(8, p)$, and the MLE of T is the bit-level formula applied to $\hat{p} = \bar{h}/8$ (the sample mean of the histogram divided by 8).

Byte match/mismatch: Setting $h = 0$ recovers the byte-level match probability:

$$\pi_T(0) = \frac{1 - s}{256} + s(1 - p)^8 = \frac{1 - s}{256} + s \left[\frac{1 + (1 - \varepsilon)^T}{2} \right]^8$$

4 Distance Estimation

4.1 From bit mismatch fraction

Given observed fraction $\hat{p}_{\text{bit}}$ of differing bits across L bytes (i.e., $8L$ bit comparisons):

$$\hat{T}_{\text{bit}} = \frac{\log(1 - 2\hat{p}_{\text{bit}})}{\log(1 - \varepsilon_{\text{eff}})} \quad (12)$$

This is always available and is closed-form.

4.2 From byte mismatch fraction

Given fraction $\hat{p}_{\text{byte}}$ of differing bytes:

$$\hat{T}_{\text{byte}} = \frac{\log(2(1 - \hat{p}_{\text{byte}})^{1/8} - 1)}{\log(1 - \varepsilon)} \quad (\text{for } r = 0) \quad (13)$$

For $r > 0$, numerical bisection on the forward model.

4.3 From the 9-state histogram (MLE)

Given observed histogram $(n_0, \dots, n_8)$ with $\sum n_h = L$:

Case $r = 0$: The distribution is $\text{Bin}(8, p)$, so $\hat{p} = \bar{h}/8$ is sufficient. The MLE is the same as the bit-level formula.

Case $r > 0$: The distribution is a 2-component binomial mixture with parameters (s, p) coupled through T . The unconstrained MLE of (s, p) can be obtained via EM:

E-step: Posterior weight for the signal component:

$$w_h = \frac{s \text{Bin}(h; 8, p)}{(1 - s) \text{Bin}(h; 8, 1/2) + s \text{Bin}(h; 8, p)}$$

M-step (closed form):

$$\hat{s} = \frac{\sum_h n_h w_h}{L} \quad (14)$$

$$\hat{p} = \frac{\sum_h h n_h w_h}{8 \sum_h n_h w_h} \quad (15)$$

After convergence, two independent distance estimates:

$$T_p = \frac{\log(1 - 2\hat{p})}{\log(1 - \varepsilon)} \quad (\text{from bit-noise accumulation}) \quad (16)$$

$$T_s = \frac{\log \hat{s}}{\log(1 - r)} \quad (\text{from byte-noise accumulation}) \quad (17)$$

Under the model, $T_p \approx T_s$. The discrepancy $|T_p - T_s|$ is a diagnostic for model violations (e.g., mutations from sources other than the noisy copy channel).

4.4 Goodness of fit

The Pearson chi-squared statistic

$$\chi^2 = \sum_{h=0}^8 \frac{(n_h - L \hat{\pi}_T(h))^2}{L \hat{\pi}_T(h)}$$

tests whether the observed histogram is consistent with the copy-noise model. Under the null hypothesis, $\chi^2 \sim \chi_d^2$ where d is the number of bins with expected count ≥ 5 , minus the number of fitted parameters.

5 Empirical Results

5.1 Three spreader variants

We compared three self-replicating programs on an 8×8 board over 2×10^6 interrupts:

Preset	Code size	Copies per rep	Noisy copies	Swaps	Avg p_{bit}
spreader	38 B	512	3,462	114,620	0.043
brk-spreader	15 B	1024 (atomic)	18,722	269,101	0.061
mini-spreader	24 B	27	8,178	213,976	0.076

Table 1: Comparison of spreader variants. “Copies per rep” is the number of bytes copied per replication event. The BRK spreader delegates copying to the engine via the noisy-copy BRK operand, which copies the entire 1024-byte cell atomically.

5.2 Histogram diagnostics

For all three spreaders, the chi-squared goodness-of-fit statistic strongly rejects the pure copy-noise model ($\chi^2 \gg$ critical value at any significance level). The observed histograms are bimodal:

- Most bytes are identical ($h = 0$): faithful copies
- A fraction are completely unrelated ($h \sim \text{Bin}(8, 1/2)$): overwrites from unrelated cells

This is consistent with a mixture model where the “noise” comes from *partial overwrites and cross-contamination* (random cells writing into each other’s code regions), not from the per-bit noise channel. The 9-state histogram model provides a natural diagnostic for detecting these non-copy mutation sources.

The BRK spreader, which uses the engine’s built-in noisy copy (operating on all 1024 bytes atomically), should in principle match the model better than the LDA/STA spreaders. In practice, cross-contamination from random cells still dominates the mutation signal.